

# Python Programming

## Unit 04 – Lecture 05: DB-API 2.0, CRUD Operations, MongoDB Overview

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Repository: <https://github.com/tali7c/Python-Programming>

# Quick Links

Core Concepts

Demo

Interactive

Summary

# Agenda

1 Core Concepts

2 Demo

3 Interactive

4 Summary

# Learning Outcomes

- Understand the DB-API 2.0 workflow (connection, cursor, execute)

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- Perform CRUD operations using parameterised queries
- Fetch results using `fetchone` / `fetchall`
- Describe how MongoDB differs from relational databases

# DB-API 2.0: The Common Pattern

```
import sqlite3
con = sqlite3.connect("app.db")
cur = con.cursor()
cur.execute("SELECT * FROM students")
rows = cur.fetchall()
con.close()
```



# CRUD

- Create: INSERT

# CRUD

- **Create:** INSERT
- **Read:** SELECT

# CRUD

- **Create:** INSERT
- **Read:** SELECT
- **Update:** UPDATE

# CRUD

- **C**reate: INSERT
- **R**ead: SELECT
- **U**ppdate: UPDATE
- **D**eleate: DELETE

# Parameterised Queries (Important)

- Avoid string concatenation in SQL

```
cur.execute(  
    "INSERT INTO students(name, sapid) VALUES (?, ?)"  
    (name, sapid)  
)
```

# Parameterised Queries (Important)

- Avoid string concatenation in SQL
- Use placeholders to prevent SQL injection and quoting bugs

```
cur.execute(  
    "INSERT INTO students(name, sapid) VALUES (?, ?)"  
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)
```

# MongoDB (High-Level View)

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- MongoDB is a document database (NoSQL)
- Stores JSON-like documents in collections
- Useful when schema is flexible or data is nested
- Python driver: `pymongo` (requires installation)

# Demo: SQLite CRUD + Optional MongoDB Stub

- SQLite CRUD: `demo/sqlite_crud_demo.py`

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- SQLite CRUD: `demo/sqlite_crud_demo.py`
- MongoDB (optional): `demo/mongodb_optional_demo.py`

# Checkpoint 1

**Question:** What is the purpose of `cursor.execute(...)`?

## Checkpoint 2

**Question:** Why are parameterised queries safer than string concatenation?

# Think-Pair-Share

Discuss:

- Which operations should be allowed in a student database app: Insert, Update, Delete? Why?

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- CRUD maps to SQL commands (INSERT/SELECT/UPDATE/DELETE)
- Parameterised queries improve safety and correctness
- MongoDB stores documents (NoSQL) and is accessed via drivers like pymongo

# Exit Question

Write one SQL query to update marks of a student with a given SAP ID.